

Ravninski lik $D \subset \mathbb{R}^2$ $\rho = \rho(x, y)$

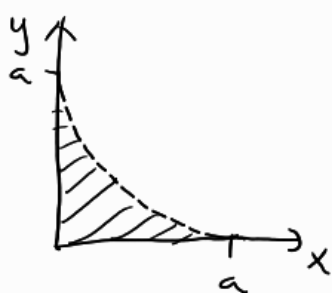
$$\bullet m = \iint_D \rho dx dy$$

$$\bullet \text{težišče } x^* = \frac{1}{m} \iint_D \rho x dx dy, \quad y^* = \frac{1}{m} \iint_D \rho y dx dy$$

$$\bullet \text{vzt. moment } J = \iint_D \rho \underbrace{(x^2 + y^2)}_{\substack{\text{kvadrat razdalje} \\ \text{do osi rotacije}}} dx dy$$

16. $x > 0, y > 0, x^{2/3} + y^{2/3} < a^{2/3}, a > 0$ fiksen, homogeno obl.

skizni:



$m, x^*, y^* = ?$

$$\begin{aligned} m &= \iint_D \rho dx dy = \rho \iint_D dx dy = \rho \int_0^a \int_0^{a(a^{2/3}-x^{2/3})^{3/2}} dy dx = \\ &= \rho \int_0^a (a^{2/3}-x^{2/3})^{3/2} dx = \rho a \int_0^1 (1-t^{2/3})^{3/2} dt = \\ &= \rho a^2 \int_0^1 (1-t^{2/3})^{3/2} dt = \frac{3}{2} \rho a^2 \int_0^1 (1-u)^{3/2} u^{1/2} du = \frac{3}{2} \rho a^2 B\left(\frac{5}{2}, \frac{3}{2}\right) = \\ &= \frac{3}{2} \rho a^2 \frac{\Gamma(\frac{5}{2})\Gamma(\frac{3}{2})}{\Gamma(4)} = \frac{3}{2} \rho a^2 \frac{\frac{3}{2} \cdot \frac{1}{2} \Gamma(\frac{1}{2})^2}{3!} = \rho a^2 \frac{3}{32} \sqrt{\pi}^2 = \underline{\underline{\frac{3}{32} \rho a^2 \pi}} \end{aligned}$$

$$\frac{x}{a} = t$$

$$\frac{1}{a} dx = dt$$

$$t^{2/3} = u$$

$$t = u^{3/2}$$

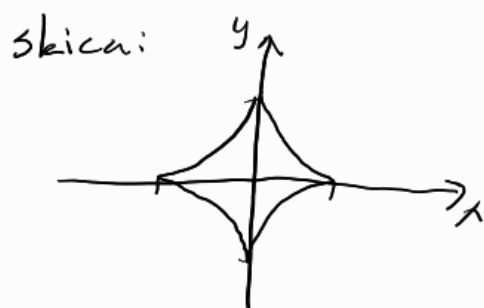
$$dt = \frac{3}{2} u^{1/2} du$$

$$x^* = \iint_D \rho x dx dy = \frac{\rho}{m} \int_0^a \int_0^{a(a^{2/3}-x^{2/3})^{3/2}} x dx dy = \frac{\rho a^3}{m} \int_0^1 (1-t^{2/3})^{3/2} t dt =$$

$$\begin{aligned} &= \frac{3 \rho a^3}{2m} \int_0^1 (1-u)^{3/2} u^2 du = \frac{3}{2} \frac{\rho a^3}{m} B\left(\frac{5}{2}, 3\right) = \frac{3}{2} \frac{\rho a^3}{m} \frac{\frac{3}{2} \Gamma(\frac{1}{2}) \Gamma(3)}{\Gamma(\frac{11}{2})} = \\ &= \frac{\rho a^3}{m} \frac{2^3}{2 \cdot 7 \cdot 5} = \frac{8}{105} \frac{\rho a^3 \cdot 32}{3 \rho a^2 \pi} = \frac{2^8}{315 \pi} a \end{aligned}$$

$$y^* = \frac{2^8}{315 \pi} a \text{ (simetrija)}$$

18. Izračunaj $\frac{J}{m}$, astroida $x^{2/3} + y^{2/3} = a^{2/3}$; a fiksen; homogen lik



$$m = 4 \cdot \frac{3 \rho a^2 \pi}{32} = \frac{3}{8} \rho a^2 \pi$$

$$J = \frac{\rho}{3} \frac{8}{a^2 \pi} \iint_D (x^2 + y^2) dx dy = 4 \cdot \frac{8}{3 \pi a^2} \iint_{x,y \geq 0} (x^2 + y^2) dx dy$$

sim. glede na x, y os

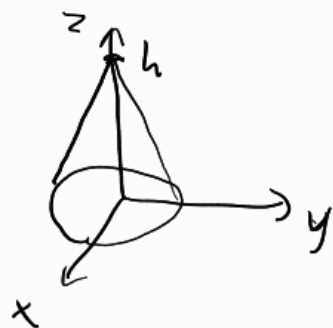
$$J = 2 \cdot \frac{32}{3 \pi a^2} \iint_D x^2 dx dy = \int_0^a x^2 (a^{2/3} - x^{2/3})^{3/2} dx =$$

simetrično
glede na y=x

$$= \frac{64}{3 \pi} a^2 \int_0^1 t^2 (1-t^{2/3})^{3/2} dt = \frac{64}{3 \pi} a^2 \cdot \frac{3}{2} \int_0^1 u^{3/2} (1-u)^{3/2} du$$

$$= \frac{4}{\pi} a^2 \frac{\frac{3}{2} \frac{3}{2} \frac{1}{2} \frac{1}{2} \frac{1}{2} \frac{1}{2}}{2 \cdot 6 \cdot 8 \cdot 4 \cdot 8 \cdot 2 \cdot 1} = \frac{7}{32} a^2$$

19. Polkrožni stožec, $\rho \propto z$. $m, T, J_z = ?$



$$\rho = kz \quad \varphi \in [0, 2\pi)$$

$$r \in [0, R - \frac{R}{h} z]$$

$$z \in [0, h]$$

$$m = \int_0^{2\pi} \int_0^h \int_0^{R - \frac{R}{h} z} \rho(z) r dr dz d\varphi = 2\pi \int_0^h \int_0^{R - \frac{R}{h} z} k z r dr dz = \pi k \int_0^h (R - \frac{R}{h} z)^2 z dz =$$

$$= \pi k R^2 \int_0^h \left(z - 2 \frac{z^2}{h} + \frac{z^3}{h^2} \right) dz = \pi k R^2 \left(\frac{h^2}{2} - 2 \frac{h^3}{3h} + \frac{h^4}{4h^2} \right) = \pi k R^2 \frac{h^2}{12}$$

$$x^*, y^* = 0$$

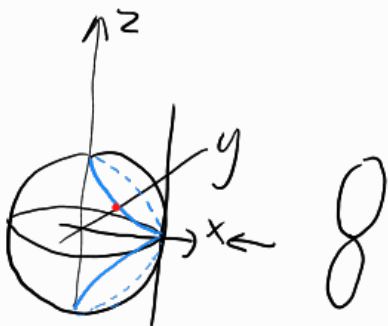
$$\dots z^* = \frac{3}{5} h \rightarrow \text{Težišče: } (0, 0, \frac{3}{5} h)$$

$$\begin{aligned}
 J_2 &= \int_0^h \int_0^{2\pi} \int_0^{R-\frac{R}{h}z} k z \cdot r \cdot r \, dr \, dz \, d\varphi = \frac{2\pi}{4} \int_0^h k z \left(R - \frac{R}{h}z\right)^4 dz = \\
 &= \frac{R^4 k \pi}{2} \int_0^h z \left(1 - \frac{z}{h}\right)^4 dz = \frac{R^4 k \pi h^2}{2} \int_0^1 t (1-t)^4 dt = \frac{R^4 k \pi h^2}{2} B(2, 5) = \\
 \frac{z}{h} &= t \\
 \frac{1}{h} dz &= dt \\
 &= \frac{R^4 k \pi h^2}{2} \frac{4 \cdot 32 \cdot 1}{65 \cdot 432} = \frac{R^4 k \pi h^2}{60} = \frac{m R^2}{5}
 \end{aligned}$$

① Parametriziraj krivuljo

$$x^2 + y^2 + z^2 = 4$$

$$(x-1)^2 + y^2 = 1, \text{ tangenta u } x=1, y \leq 0, z \geq 0.$$



$$y(x) = \pm \sqrt{1 - (x^2 - 2x + 1)} = \pm \sqrt{2x - x^2}$$

$$z(x) = \pm \sqrt{4 - x^2 - (2x - x^2)} = \pm \sqrt{4 - 2x}$$

$$x \in [0, 2]$$

$$\vec{r}(x) = \begin{bmatrix} x \\ \pm \sqrt{2x - x^2} \\ \pm \sqrt{4 - 2x} \end{bmatrix}$$

$$\vec{r}(x=1) = \begin{bmatrix} 1 \\ -1 \\ \sqrt{2} \end{bmatrix}$$

$$\dot{\vec{r}} = \begin{bmatrix} 1 \\ \pm \frac{1}{2}(2x - x^2)^{-\frac{1}{2}}(2 - 2x) \\ \pm \frac{1}{2}(4 - 2x)^{-\frac{1}{2}}(-2) \end{bmatrix} = \begin{bmatrix} 1 \\ \mp (2x - x^2)^{-\frac{1}{2}}(x-1) \\ \mp (4 - 2x)^{-\frac{1}{2}} \end{bmatrix}$$

$$\dot{\vec{r}}(x=1) = \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \rightarrow R = \vec{r}(1) + s \cdot \dot{\vec{r}}(1) = \begin{bmatrix} 1 \\ -1 \\ \sqrt{2} \end{bmatrix} + s \cdot \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$